P P SAVANI UNIVERSITY · SCHOOL OF ENGINEERING

PromptForm AI

AI-Powered Form Generation Platform

Internship / UDP / Training Final Project · Academic Year 2026–2027

School of Engineering

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PROGRAMME
B.Tech Computer Science & Engineering · Sem VII

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ORGANISATION
OCCUBIT SOLUTION

01 Abstract

PromptForm AI turns a one-sentence prompt or an uploaded document into a complete, validated and shareable online form in seconds. Google Gemini, guided by rule-based AIFormBrain checks, builds the form; users then edit, publish, run quizzes and analyse responses on one platform.

02 Problem Statement

- **Manual:** every field and rule set by hand
- **Slow:** 15–25 min for a 10-field form
- **Error-prone:** wrong types, re-typed PDFs
- **Fragmented:** separate quiz & analytics tools

03 Objectives

- Prompt → complete form in under 10 s
- Digitise PDF, DOCX, XLSX and image forms
- 10+ field types with automatic validation
- Graded quizzes, analytics and export
- Secure, team-ready cloud deployment

04 Technology Stack

Frontend	Next.js · React · TypeScript
Backend	Node.js · Express.js
Database	PostgreSQL · Prisma ORM
AI	Google Gemini API

05 Proposed Solution

Describe a form in plain English or upload an existing one. PromptForm AI plans the fields, generates a typed and validated form, and opens it ready to edit and share.

PROMPT
"Patient intake form for a dental clinic with contact details, medical history and appointment date"
or upload PDF · DOCX · XLSX · image

→

AIFORMBRAIN + GEMINI
1. Detect intent & domain
2. Plan typed fields & rules
3. Schema JSON + quality review

→

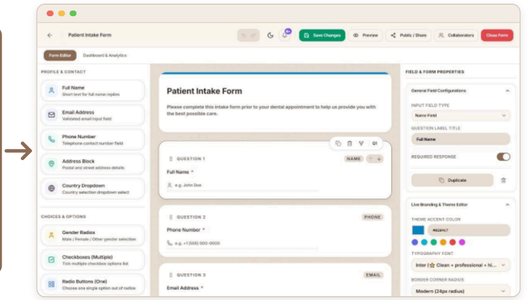


Figure 1. In the final demo, this prompt produced a six-question Patient Intake Form (visual builder).

06 System Architecture



Data: Express.js → Prisma ORM → PostgreSQL · **Hosting:** Vercel (frontend), Render (backend)

Figure 2. Request flow from prompt to interactive form.

07 Workflow & Methodology

1	2	3	4	5	6
Sign in	**Describe**	**Generate**	**Customise**	**Publish**	**Analyse**
e-mail or Google account	type a prompt or upload a file	AI builds typed fields	edit questions and theme	share a unique short link	responses and CSV export

Methodology: Agile Scrum with two-week sprints over 14 weeks and weekly mentor reviews; validated with 50 automated test cases.

08 Key Features

- **AI form generation**
- **Live preview**
- **Field validation**
- **Secure sign-in**
- **Team collaboration**
- **Editable forms**
- **11 question types**
- **Quizzes & anti-cheat**
- **Analytics & export**

09 Results & Benefits

50/50	≈ 8 s	10/10
tests passed	prompt to form	objectives met

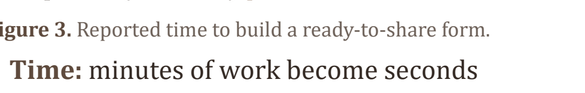


Figure 3. Reported time to build a ready-to-share form.

- **Time:** minutes of work become seconds
- **Ease:** types and validation set automatically
- **Usability:** live preview and 3 display modes

10 Conclusion & Future Scope

- **Conclusion:** a deployed, secure platform that builds validated forms in seconds.
- **Applications:** registrations, patient intake, surveys and quizzes.
- **Limitations:** relies on the Gemini API; 30–50 s cold start on free hosting.
- **Future:** payments, AI conditional logic, multilingual input.

References

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Scan to Visit PromptForm AI
promptform-ai-frontend.vercel.app